CI/CD Workflow Report

GitHub + Jenkins Declarative Pipeline

Project	git-task
GitHub Repository	Ar-Sawan/git-task
Branch	master
CI/CD Tool	Jenkins
Pipeline File	Jenkinsfile
Pipeline	Build → Test → Deploy

1. Objective

The objective is to prepare a CI/CD workflow using GitHub and Jenkins, define pipeline stages, execute the workflow automatically, and document the results with supporting screenshots.

2. Pipeline Architecture

Local Git / Developer	GitHub	Jenkins	Pipeline	Result
Jenkinsfile committed locally	master branch source repository	GitHub-CI-CD Jenkins Job	Build ↓ Test ↓ Deploy	SUCCESS

Workflow: Local Git repository → GitHub master branch → Jenkins checkout → Build → Test → Deploy → successful pipeline.

3. Pipeline Stages

Stage	Purpose	Observed Result
Build	Represents the application build step.	Build successful
Test	Represents automated testing after build.	Tests passed
Deploy	Represents deployment after successful testing.	Deploy successful
Post-success	Confirms successful completion of the pipeline.	Pipeline completed successfully

4. Automation Results

Jenkins successfully retrieved the Jenkinsfile from GitHub and executed the declarative pipeline. The Build, Test, and Deploy stages completed successfully, and Jenkins reported the overall result as SUCCESS.

Verification	Result
GitHub repository checkout	Successful

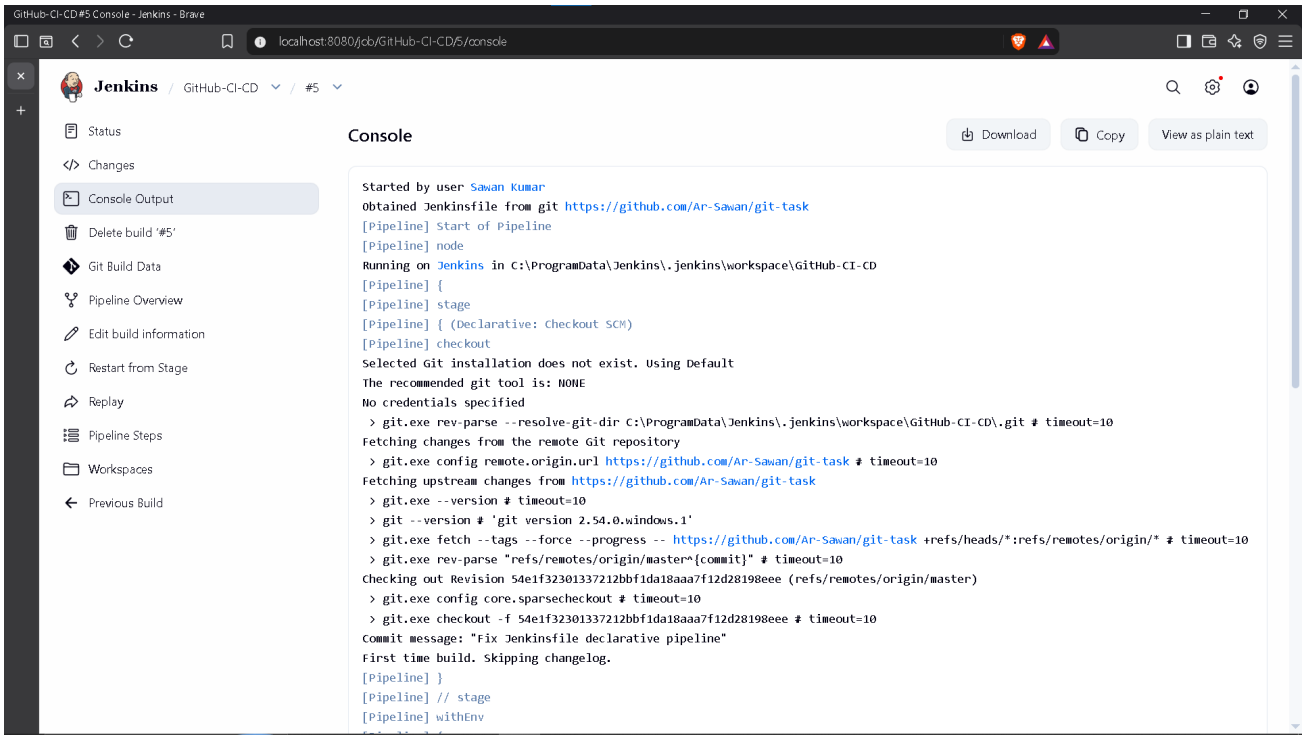
Jenkinsfile execution	Successful
Build stage	Successful
Test stage	Successful
Deploy stage	Successful
Overall pipeline	SUCCESS

5. Jenkinsfile Definition

```
pipeline {
    agent any
    stages {
        stage('Build') { steps { echo 'Build successful' } }
        stage('Test')  { steps { echo 'Tests passed' } }
        stage('Deploy'){ steps { echo 'Deploy successful' } }
    }
    post {
        success { echo 'Pipeline completed successfully!' }
    }
}
```

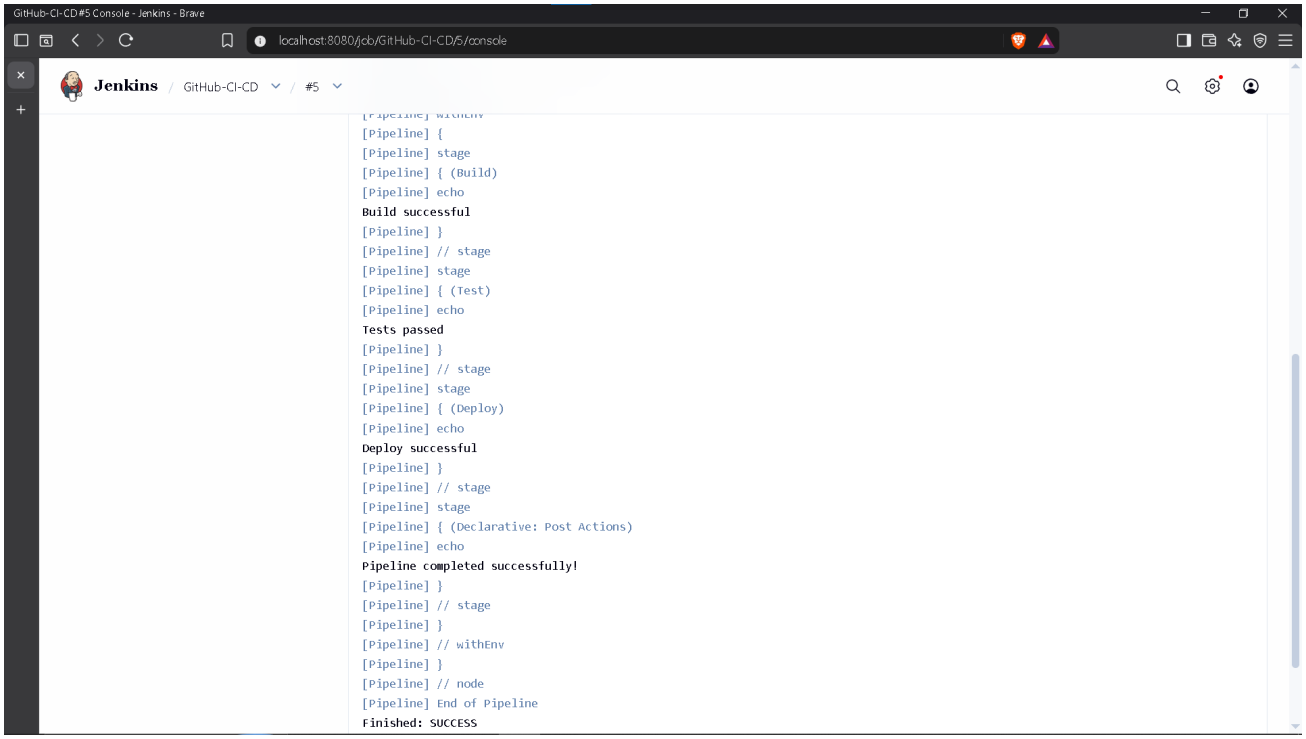
6. Supporting Screenshots

Screenshot 1 – Jenkins successful pipeline execution



Supporting evidence captured during the practical.

Screenshot 2 – Git push / GitHub integration evidence



Supporting evidence captured during the practical.

7. Observations

• GitHub is used as the source-control repository for the Jenkins pipeline.

• Jenkins retrieves the Jenkinsfile from the master branch.

• The workflow is organized into Build, Test, and Deploy stages.

• The console output confirms successful execution of all three stages.

• The final Jenkins result is shown as SUCCESS.

8. Conclusion

The CI/CD workflow was implemented and successfully executed using GitHub and Jenkins. The pipeline automates the Build, Test, and Deploy sequence, and the screenshots provide evidence of GitHub integration and successful Jenkins execution.